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Trustworthy AI in Wound Care Documentation

Oskar Gustafsson OSKAR.GUSTAFSSON@MOLNLYCKE.COM
Halmstad University, Sweden
Mölnlycke Health Care, Sweden

Jens Lundström JENS.R.LUNDSTROM@HH.SE
Halmstad University, Sweden

Ulf Johansson ULF.JOHANSSON@JU.SE
Dept. of Computing, Jönköping University, Sweden

John Pavia JOHN.PAVIA@MOLNLYCKE.COM
Mölnlycke Health Care, Sweden

Niclas Ståhl NICLAS.STAHL@JU.SE
Dept. of Computing, Jönköping University, Sweden

Cecilia Sönströd CECILIA.SONSTROD@JU.SE
Dept. of Computing, Jönköping University, Sweden

Ernst Ahlberg ERNST.AHLBERG@MOLNLYCKE.COM
Royal Holloway University, United Kingdom
Mölnlycke Health Care, Sweden

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Abstract

Efficient and trustworthy support for clinical documentation is becoming increasingly important as reporting requirements expand and EHR systems continue to increase in complexity. In wound care, where highly detailed structured assessments are a routine part of practice, there is a particular need for methods that assist clinicians through reliable recommendations. AI-based tools offer promising opportunities to reduce the documentation burden, but their value ultimately depends on the trustworthiness of the guidance they provide, as their recommendations feed directly into patient care and outcomes. In this work, we build on a unified uncertainty-quantification framework that combines Conformal Prediction with Venn–Abers predictors and extend it by integrating Shapley-value explanations into the same workflow. The resulting system produces prediction sets that narrow the label space for structured wound-care documentation while providing statistical coverage guarantees, calibrated inclusion probabilities, and interpretable explanations for the associated confidence. In practice, this kind of system could reduce documentation time and give clinicians a straightforward view of both the model's uncertainty and the reasons behind its recommendations.

Keywords: Trustworthy AI, Uncertainty Quantification, Conformal prediction, Explainable AI (XAI), Wound Care, AI in Wound Care, AI in Healthcare

1. Introduction

The burden of structured clinical documentation has increased substantially with the adoption of electronic health record (EHR) systems and has been identified as a major contributor to clinician workload and burnout Murad et al. (2024). This trend is driven in part by the growing demand for more detailed and structured data to support quality monitoring and emerging AI-driven tools Rajkomar et al. (2019); Beam and Kohane (2018); Jensen et al. (2012). Many healthcare professionals report that the time and cognitive effort required for accurate record-keeping detracts from patient care American Medical Informatics Association (2024). This perception is supported by quantitative evidence: time-motion studies show that clinicians spend nearly twice as much time on EHR tasks as on direct patient care Arndt et al. (2017). In practice, this means that documentation demands do not only exhaust clinicians, but also erode the time available for meaningful patient interaction.

In wound care specifically, clinicians require substantial time and attention for documentation because assessments are highly detailed and visits occur on a regular schedule Irestig et al. (2025). Because clinicians must prioritize patients' immediate health needs, wound care documentation is not always given precedence. Gustafsson et al. (2026a) demonstrate that, in an integrated platform of data from hospitals, primary care and acute care in Region Halland in Sweden, the ICD-10 code for "unspecified wound" is used more frequently than all chronic wound categories—pressure ulcers, diabetic ulcers, and leg ulcers—combined. In the dataset used in this study, clinicians must choose among 16 distinct Periwound Skin types and 16 different wound edge types. While clinically meaningful, this level of detail creates a substantial cognitive load, especially in home-care settings where documentation is often performed on small mobile devices under time pressure.

In a systematic review of AI-powered documentation systems Bracken et al. (2025) describe that clinicians are cautiously optimistic about the implementation of AI-assisted documentation in healthcare. Clinicians' primary concern centered on the reliability and validity of recommendations generated by AI-based decision-support systems, reflecting apprehension about the consistency, accuracy, and evidentiary grounding of model outputs. Further, Lindberg et al. (2026) noted in their preliminary analysis that a recurring theme among clinicians in wound care was the importance of understanding the rationale behind AI-generated recommendations. Karnehed et al. (2025) report that clinicians in wound care in municipal home healthcare raises concerns that the integration of AI-driven automation might overlook the embodied practices integral to wound care such as nuanced judgement of sensory inputs such as smell and touch. Further, concerns were raised about the loss of the relational aspect of wound care. In the deployment of AI-models, it is also important to keep in mind the *"garbage in - garbage out"* principle, namely that a model can never be better than the data it is trained on Kilkenny and Robinson (2018). EHR systems typically have quality issues across a variety of domains such as completeness, correctness, concordance and plausibility Lewis et al. (2023). Further, Horsky et al. (2012) describe that the use of drop-down menus in healthcare makes clinical work cognitively demanding and error-prone. Similarly, Karnehed et al. (2025) describe that clinicians feel that existing digital systems increase *"administrative burdens rather than streamlining care"*. The overall picture reflects a persistent tension between the demand for structured, high-quality clinical documentation and the practical constraints of real-world workflows.

In this work, we address this tension by integrating AI-based decision support, utilizing techniques from uncertainty quantification and explainable AI, into a wound care documentation tool. The aim is to reduce the decision space presented to the clinician, i.e„ the number of choices that must be made during documentation, in a trustworthy manner. Technically, the solution leverages model uncertainty to collapse or defer low-confidence options in the tool, by constructing *marginally valid set predictions* Angelopoulos and Bates (2021); Vovk et al. (2022) , with conformal prediction. Specifically, the prediction set determines the options shown in the tool. Additionally, we assign a well-calibrated probability of the true label being in the set of presented options and suggest a way of reducing the cognitive load of processing this information, by binning probabilities into a small number of interpretable categories. Lastly, SHAP, a method within explainable AI, is used to provide justifications for the confidence in the suggested set of options, supporting trust and enabling clinicians to quickly validate or override suggestions. Together, these components constitute a documentation-support system intended to assist clinical documentation by providing statistical guarantees around model-assisted decisions. Rather than presenting clinicians with extensive sets of documentation options, the system narrows the set of candidate labels while making explicit the underlying model uncertainty and the AI-based reasoning that informed each recommendation. This design has the potential to support efficient documentation and improve the transparency of model-assisted recommendations, although its effects on usability, clinical workflows, and user trust remain to be evaluated.

The paper is organized as follows. Section 2 provides the necessary technical background, including an overview of conformal prediction, Venn–Abers predictors, and a recent framework that integrates the two. It also introduces the concept of explanations in AI and presents SHAP as a method for model interpretability. Section 3 describes the methodological approach, beginning with a presentation of the data, the feature-selection and feature-engineering procedures, and an overview of the target variables. This is followed by a description of how the technical frameworks are applied to the classification problems and integrated into a documentation-support system. The section concludes with a detailed account of the experimental workflow. Section 4 outlines the results, including set-size reduction achieved through conformal prediction across classification tasks and -values, calibration plots, and SHAP waterfall plots for selected predictions. Section 5 provides a discussion of the findings, including their utility, limitations, and implications for future work. Finally, Section 6 concludes the paper.

2. Background

2.1. Conformal Prediction for Classification

Conformal prediction (CP) is a distribution-free framework for constructing prediction sets with finite-sample validity guarantees. Consider a classification problem with an input space $\mathcal{X}$ and label space $\mathcal{Y} = \{1, \ldots, C\}$. Let $(X_i, Y_i)_{i=1}^n$ be exchangeable examples, typically assumed to be i.i.d.

The goal of conformal prediction is to construct, for a new test object X_{n+1}, a prediction set

$$\Gamma_\varepsilon(X_{n+1}) \subseteq \mathcal{Y}$$

such that the following coverage guarantee holds:

$$\mathbb{P}\left(Y_{n+1} \in \Gamma_\varepsilon(X_{n+1})\right) \geq 1 - \varepsilon, \tag{1}$$

for a user-specified significance level $\varepsilon \in (0, 1)$.

Let $A : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$ be a nonconformity function, where larger values indicate that a pair (x, y) is less conforming to the training data according to the underlying predictive algorithm and the chosen nonconformity measure. Typically, A is derived from an underlying model, e.g., based on class probabilities or margins.

In the inductive (split) conformal setting, the data is partitioned into a proper training set and a calibration set $\mathcal{C} = \{(X_i, Y_i)\}_{i=1}^m$. The model is trained on the proper training set, and nonconformity scores are computed on the calibration set:

$$\alpha_i = A(X_i, Y_i), \quad i = 1, \ldots, m.$$

For a test object x, we evaluate each candidate label $y \in \mathcal{Y}$ by computing

$$\alpha_{n+1}(y) = A(x, y).$$

For each label $y \in \mathcal{Y}$, a conformal p-value is defined as

$$p(y) = \frac{|\{i = 1, \ldots, m : \alpha_i \geq \alpha_{n+1}(y)\}| + 1}{m + 1}.$$

The conformal prediction set is then given by

$$\Gamma_\varepsilon(x) = \{y \in \mathcal{Y} : p(y) > \varepsilon\}.$$

Under the assumption that the calibration examples and the test example are exchangeable, conformal prediction satisfies the finite-sample validity property in equation 1. This guarantee holds regardless of the choice of underlying model or nonconformity function, and does not rely on distributional assumptions.

In the multi-class setting ($C \geq 2$), the prediction set $\Gamma_\varepsilon(x)$ may contain any number of labels:

$$|\Gamma_\varepsilon(x)| \in \{0, 1, \ldots, C\}.$$

In particular,

- $|\Gamma_\varepsilon(x)| = 1$ corresponds to a confident, singleton prediction,
- $|\Gamma_\varepsilon(x)| > 1$ reflects ambiguity among several plausible labels,
- $|\Gamma_\varepsilon(x)| = 0$ indicates that all labels are rejected at level ε.

The occurrence of empty prediction sets is theoretically possible, although typically rare in well-calibrated settings.

An alternative and often insightful way to interpret conformal prediction in the multi-class setting is as a *label rejection mechanism*. Rather than selecting a single best label, CP evaluates each label $y \in \mathcal{Y}$ individually and *rejects* those that are insufficiently supported by the data.

Specifically, a label y is excluded from $\Gamma_\varepsilon(x)$ if its p-value satisfies $p(y) \leq \varepsilon$, meaning that it is among the most nonconforming labels relative to the calibration data. The resulting prediction set can thus be written as

$$\Gamma_\varepsilon(x) = \mathcal{Y} \setminus \{y \in \mathcal{Y} : p(y) \leq \varepsilon\}.$$

From this perspective, conformal prediction does not directly aim to identify the true label, but instead to eliminate labels that are unlikely. The remaining set $\Gamma_\varepsilon(x)$ consists of labels that are *not rejected* at significance level ε.

The validity guarantee then implies that, with probability at least $1 - \varepsilon$, the true label is not among the rejected labels. Equivalently, the true label is contained in the retained set $\Gamma_\varepsilon(x)$ with high probability.

This interpretation highlights that conformal prediction provides a principled, statistically valid way of narrowing down the set of plausible labels, where the size of the retained set adapts to the difficulty of the instance.

2.2. Alternative Conformal Prediction Methods: APS & RAPS

Adaptive Prediction Sets (APS) is a conformal prediction method for multiclass classification that constructs prediction sets by accumulating class probabilities in descending order until a specified threshold is reached. Unlike standard conformal prediction, which treats all classes equally, APS takes the probability distribution produced by the classifier into account, often resulting in smaller and more informative prediction sets while maintaining the desired coverage guarantee.

Regularized Adaptive Prediction Sets (RAPS) extends APS by introducing a regularization term that penalizes the inclusion of lower-ranked classes. This modification reduces the tendency of APS to produce unnecessarily large prediction sets when many classes have small but non-negligible probabilities. Consequently, RAPS often achieves greater efficiency, i.e. smaller prediction sets, while retaining the finite-sample coverage guarantees provided by conformal prediction.

2.3. Venn-Abers Predictors

Venn–Abers (VA) predictors are a class of multi-probabilistic predictors designed for binary classification, providing well-calibrated probability estimates with strong validity guarantees. Let $\mathcal{Y} = \{0, 1\}$ and assume exchangeable data $(X_i, Y_i)_{i=1}^n$.

Unlike standard probabilistic classifiers that output a single probability estimate, VA predictors produce a *pair* of probabilities for a test instance, reflecting uncertainty in the prediction.

Assume a scoring function $s : \mathcal{X} \to \mathbb{R}$, obtained from a trained classification model, where higher values indicate stronger evidence for class 1. The VA framework applies isotonic regression to calibrate these scores.

Let $\mathcal{C} = \{(X_i, Y_i)\}_{i=1}^m$ be a calibration set. Compute scores

$$s_i = s(X_i), \quad i = 1, \ldots, m.$$

For a test object x, compute $s_{n+1} = s(x)$. The key idea of VA is to consider two hypothetical label assignments:

- Assign $Y_{n+1} = 0$, augment the calibration set with $(s_{n+1}, 0)$, and fit an isotonic regression model g_0.
- Assign $Y_{n+1} = 1$, augment the calibration set with $(s_{n+1}, 1)$, and fit an isotonic regression model g_1.

This results in two calibrated probability estimates:

$$p_0 = g_0(s_{n+1}), \quad p_1 = g_1(s_{n+1}).$$

The Venn–Abers predictor outputs the pair (p_0, p_1), which constitutes a *multiprobabilistic prediction*. Under the assumption of exchangeability, Venn predictors (including VA) satisfy a strong calibration property: there exists at least one probability value, p, within the multiprobabilistic prediction that is perfectly calibrated in the following sense:

$$\mathbb{E}\big[Y_{n+1} \mid P = p\big] = p \quad \text{a.s.} \tag{2}$$

The pair (p_0, p_1) is often interpreted as defining a probability interval. This interval can be viewed as capturing epistemic uncertainty about the true class probability. In particular:

- A narrow interval indicates high confidence in the probability estimate,
- A wide interval reflects uncertainty due to limited or ambiguous calibration data.

It is important to note, however, that the VA framework does not inherently define this interval interpretation; rather, it produces two discrete candidate probabilities corresponding to the two hypothetical label assignments. While the interval interpretation is convenient, the underlying theory is based on the existence of a valid selector within the multiprobabilistic output.

In many applications, Venn–Abers predictors are used as a post-hoc calibration method for an underlying classifier. In such cases, it is common to reduce the multiprobabilistic prediction to a single probability value.

A standard approach is to minimize the regret (log loss) by using

$$\hat{p} = \frac{p_1}{1 - p_0 + p_1}. \tag{3}$$

i.e., a regularized value in which the estimate is moved towards 0.5

Gustafsson et al. (2026b) introduces a self-consistent scoring rule that combines the two Venn-Abers probabilities p_0 and p_1 into a single prediction probability. The rule weights p_0 and p_1 according to the likelihood of the respective events they postulate, yielding

$$\hat{p} = \frac{p_0}{1 - p_1 + p_0}. \tag{4}$$

In this work, we adopt this self-consistent scoring rule in place of the min–max aggregation in Equation 3. A point estimate is often preferred in practice because of its simplicity, even though it discards the validity guarantees. In effect, the resulting predictor behaves similarly to standard calibration techniques—such as Platt scaling or isotonic regression—but it no longer preserves the full theoretical guarantees offered by the Venn prediction framework.

2.4. Well-Calibrated Probabilities for Set Predictions

Gustafsson et al. (2026b) introduced a method that assigns well-calibrated probabilities to set predictions. Their central observation is that a set prediction $\Gamma_\varepsilon(x)$ can be conceived of as a binary split of the label space and thus be assigned a prediction probability with a Venn–Abers machine V. Within such a Venn–Abers machine, the two possible outcomes correspond to the events $y_i \in \Gamma_\varepsilon(x_i)$ and $y_i \notin \Gamma_\varepsilon(x_i)$. The expression $y_i \in \Gamma_\varepsilon(x_i)$ is thus a Bernoulli random variable.

Let O denote an *oracle selector*, that is, a selector which always chooses the probability corresponding to the correct postulate produced by V. Concretely, if $V(x_i)$ outputs the pair (p_0, p_1), then $O(V(x_i))$ is the *realized* choice between the two prediction probabilities.

With this interpretation, O satisfies

$$O(V(x_i)) = \mathbb{E}(y_i \in \Gamma_\varepsilon(x_i) \mid O(V(x_i))) \quad \text{a.s.}, \tag{5}$$

which is the analogue of equation 2, specialized to the construction in which the binary output encodes whether the true label is contained in the prediction set.

For practical purposes, the scoring rule in equation 4 collapses the multi-probabilistic prediction into a single-valued prediction probability. As described in section 2.3, this removes the distinctive theoretical property guaranteeing *perfect calibration*. However, when the interval between p_0 and p_1 is small—which is typically the case with large calibration sets—both quantities approximately satisfy 5 and are therefore *well-calibrated* Vovk and Petej (2012). The method assigns, for each set prediction made with conformal prediction, a corresponding Venn-Abers machine which provides a probability of the label being inside the set. Importantly, if the interval between p_0 and p_1 is small, the probability is well-calibrated within an equivalence class implicitly induced by the underlying model's scores, rather than explicitly chosen by the user. This represents a key distinction from Mondrian conformal prediction, where the partitioning of the feature space into equivalence classes must be specified a priori by the practitioner Vovk et al. (2022).

It should be noted that the procedure requires, in addition to a proper training set and a test set, two calibration sets; $\mathcal{C}_\mathcal{C} = \{(X_i, Y_i)\}_{i=1}^{k}$ and $\mathcal{C}_\mathcal{V} = \{(X_i, Y_i)\}_{i=k+1}^{m}$ for Conformal Prediction and Venn-Abers respectively.

2.5. Explanations

As machine learning becomes widely adopted, especially in the domain of healthcare, it is of crucial importance to maintain a high level of trust in systems and processes affected by AI. Being able to explain model average behavior (global explainability) or model explanations for specific inputs (local explainability) allows developers to take steps towards the concept known as *Trustworthy AI*. Legislation in the EU for AI-driven systems (EU's AI Act European Parliament and the Council of the European Union (2024)) operationalizes safety and trustworthiness for high-risk systems (which is a classification often linked to AI-driven applications in healthcare) by regulating transparency and human oversight among various systemic requirements. The AI Act does not enforce a particular explainable AI method for black-box machine learning models that are not inherently explainable, leaving developers to decide on a suitable implementation. For this study we have chosen to focus on local explainability of prediction sets using SHAP.

2.5.1. SHAP

The Shapley value is a solution concept from cooperative game theory that fairly attributes each contributor's marginal impact on the final outcome Shapley (1953). In machine learning, this idea is adapted into SHAP values, which quantify how much each feature pushes a model's prediction up or down. The key strength of Shapley-based methods is fairness: they compute a feature's average marginal contribution across all possible combinations of features, ensuring a consistent and principled explanation of model behavior.

This idea is formalized by the Shapley value formula:

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(|N| - |S| - 1)!}{|N|!}\,(v(S \cup \{i\}) - v(S)) \tag{6}$$

Where each coalition S represents a subset of features acting together, the term $v(S \cup \{i\}) - v(S)$ captures the marginal contribution of feature i when added to that subset, and the factorial weight $\frac{|S|!\,(|N|-|S|-1)!}{|N|!}$ ensures that this contribution is averaged fairly over all possible orders in which features can join the coalition Lundberg and Lee (2017).

Building on the Shapley value formulation in Equation 6, Kernel SHAP provides a practical, model-agnostic procedure for estimating feature attributions when the exact computation of all $2^{|N|}$ coalitions is infeasible. The method reframes the explanation task as a weighted linear regression problem over simplified binary feature representations Molnar et al. (2021). In its standard form, Kernel SHAP estimates feature contributions using marginal feature distributions and therefore does not explicitly account for dependencies among input variables. Consequently, when features are correlated, the resulting attributions should be interpreted with appropriate caution, as the allocation of importance among related features may be affected.

In the machine learning literature, SHAP is commonly employed to generate both *local* and *global* explanations of model behavior. In the context of this study, our primary objective is to assess the trustworthiness of the AI model from a clinician's perspective. Consequently, we focus exclusively on local SHAP explanations, as they offer case-specific transparency. By presenting SHAP values alongside each model suggestion, clinicians gain explicit information about the factors influencing the model's decision, thereby supporting more informed and trustworthy human–AI interaction.

SHAP is designed for binary problems. In multiclass settings, the typical approach is a one-vs-rest strategy. This approach has been critiqued because it does not take into account the dynamical relationship between classes Noé et al. (2025); Franceschi et al. (2023). A critique of this approach is the implicit assumption that each class output is a separate scalar prediction. In reality, model outputs for multiclass problems form a probability vector that lies on a simplex hyperplane. In other words, the class probabilities need to be non-negative and sum to one. The one-vs-rest approach thus fails to account for the inherent dependency structure between class probabilities. Ignoring this coupling may create misleading explanations.

Feature	Possible values	Five most frequent values
Tissue Type	6	Granulation (54%) Slough (36%) Necrotic (7%) Epithelialization (2%) Closed (1%)
Periwound Skin 1	16	Normal (45%) Edema (18%) Poorly Vascularized (12%) Flakiness/Dryness (11%) Erythema (5%)
Periwound Skin 2	15	None (88%) Edema (4%) Flakiness/Dryness (3%) Erythema (1%) Poorly vascularized skin (1%)
Edge 1	16	Normal (50%) Irregular (10%) Undermining (9%) Formation of crust (7%) Callus (6%)
Edge 2	15	None (92%) Undermining (1%) Formation of crust (1%) Maceration (1%) Epithelialization (1%)

Table 1: Overview of most common labels for the 5 target variables.

3. Method

3.1. Data

The dataset, referred to as *Wound Monitor*, used in this study was collected during home-care visits conducted by specialized wound-care clinicians in the Netherlands. In total, the cohort comprises approximately 630,000 recorded visits. Of these, slightly more than 185,000 visits include both a clinical photograph and an associated polygon annotation delineating the wound boundary. The observations correspond to 23,000 unique wounds and 13,000 unique patients. Because the original dataset was in Dutch, all variable names were translated into English. To ensure consistency, each unique variable name was translated only once: a Python dictionary stored previously generated translations, and the LLM was queried only when a new, unseen name appeared. The dataset is described in detail by Gustafsson et al. (2026a).

For each visit and each wound, clinicians routinely documented five categorical variables: Tissue Type, Periwound Skin 1, Periwound Skin 2, Edge 1, and Edge 2. Tissue Type describes the predominant tissue present in the wound bed. Periwound Skin captures the condition of the skin immediately surrounding the wound. Edge describes the characteristics of the wound margins. These variables constitute the classification targets examined in this study. Table 1 summarizes the distribution of the most prevalent labels across the five classification tasks as well as the number of classes within each task after removing every class with less than 100 observations.

Periwound Skin 2 and Edge 2 are documented only when clinicians judge that more than one descriptor is required to describe the respective conditions. As a result, the label *None* is by far the most frequent category for these two variables. We chose to include these labels for the prediction task, as it is practically and clinically relevant to determine that a single descriptor is sufficient to describe the respective conditions. For Tissue Type, Periwound Skin 1, and Edge 1, the most common label constitutes approximately 50% of all observations. Overall, the distribution of labels is highly imbalanced.

3.1.1. Feature selection and engineering

Below, we describe the selection and engineering of the features used for the five classification tasks. Twenty-six features were taken directly from the raw dataset. These correspond to anatomical location (8 features), wound type (11 features) and gradings of wound complexity (7 features). The complexity gradings are based on several established classification systems, each applied depending on the specific wound type. A detailed description of which systems were used can be found in Gustafsson et al. (2026a).

In a clinical setting, clinicians typically have access to the previously documented history of a wound. To leverage this information, we constructed temporal features using exponentially decayed recurrence relations, as formalized in Equation 7. This approach enables the model to integrate previously documented wound-history data while weighting recent observations more heavily than distant ones.

$$h_0^{(a)} = 0, \qquad h_n^{(a)} = \begin{cases} 1 + 0.5\, h_{n-1}^{(a)}, & \text{if } y_{n-1} = a, \\ 0.5\, h_{n-1}^{(a)}, & \text{otherwise.} \end{cases} \tag{7}$$

A notable property of Equation 7 is that, for all visits after the first, precisely one history feature $h_n^{(a)}$ attains a value of 1 or greater, namely the feature associated with the label observed at the previous time step. This gives a clear interpretation of the feature. Another notable property is that no historical feature can be equal to or exceed 2. A value, $h_n^{(a)} \approx 2$, indicates a long history of repeated observations of the same label a.

The images were annotated with polygons by clinicians. For each image, the pixels inside the polygon were labeled as one of seven predefined colours based on their nearest neighbour in the CIE L*a*b* colour space, a perceptually uniform representation in which L^* denotes lightness and a^* and b^* encode chromaticity along the green–red and blue–yellow axes. The translation from RGB to CIE L*a*b* makes nearest-neighbour assignment more robust: colours that appear similar to clinicians are also close in the feature space, reducing misclassification due to illumination changes or device-specific colour profiles ISO (2019). Moreover, separating lightness L from chromaticity a, b, allows the method to focus on hue-related differences that are clinically relevant while being less affected by shadows or brightness variations. With this procedure, seven features were created, each corresponding to the proportion of pixels assigned to the respective colours.

Each image included a ruler with two annotated points indicating a specified distance—typically 1 cm. This scale reference was used to define the regions around the boundary of the polygon corresponding to the wound Edge and the Periwound Skin, enabling the proportional distribution of colour categories to be computed for these regions, resulting in 14 additional features. The term *Periwound Skin* lacks a universally accepted definition in the clinical literature; guidelines typically describe it as the skin immediately adjacent to the wound Edge, often within approximately 1–4 cm Wound, Ostomy and Continence Nurses Society (2016); National Pressure Ulcer Advisory Panel et al. (2014). For feature extraction, we operationally defined the periwound region as -1 to +3 cm around the boundary defined by the polygon. The wound Edge was defined as the boundary between the wound bed and the surrounding intact skin, consistent with established wound assessment guidelines Wound, Ostomy and Continence Nurses Society (2012). Although clinical sources do not specify whether this Edge should have a measurable width, for the purposes of feature engineering we operationally defined the wound Edge as a 1 cm band on either side of the boundary delineated by the clinician-annotated polygon.

Lastly, the geometry of the polygon was utilized to calculate the perimeter, the area, and three irregularity metrics; $\frac{area}{perimeter^2}$, $\frac{\text{area}}{\text{area of convex hull}}$ and $\frac{\text{perimeter}}{\text{perimeter of convex hull}}$.

3.2. Application

The purpose of this article is to reduce the burden of documentation for clinicians in a trustworthy way. This is achieved with three main steps.

- **Construct marginally valid set predictions** with conformal prediction.
- **Assign well-calibrated prediction probabilities** of the label being in the set with IVAP.
- **Provide local feature level explanations for the confidence in the set** with SHAP.

Following the formulation in Gustafsson et al. (2026b), we treat a prediction set as inducing a binary classification problem:

Is the true label contained in the prediction set?

This reframing avoids the difficulties described in 2.4.1 of applying SHAP directly to multiclass probability vectors, which lie on a simplex and exhibit inherent dependency between class probabilities. Instead, the binary formulation yields a single scalar target suitable for Shapley analysis. This means the calibrated output of the Venn-Abers functions as the payoff function $v(S)$ in 6.

The integration of these three approaches produces an AI-recommendation system that narrows the set of clinical alternatives in a trustworthy manner. Instead of presenting clinicians with a cognitively demanding drop-down menu, the system highlights only a small number of high-probability options, while making both uncertainty and model reasoning transparent. To further avoid adding cognitive load, we propose reducing the granularity of the probability outputs by mapping each prediction to one of five probability intervals: 0–50% (low), 50–75% (moderate), 75–90% (medium), 90–98% (high), and 98–100% (very high). These thresholds were chosen arbitrarily. In a real-world implementation, the thresholds need to be decided in conversation with clinicians.

Figure 1 illustrates a potential phone-based interface of the framework, demonstrating how instead of a dropdown menu, a clinician can interact with a reduced set of options. Further, Figure 1 demonstrates how the binned probabilities can be presented in an interpretable format. The lower panel illustrates how the outputs of an explainable AI (XAI) technique, such as Shapley values, can be abstracted into a representation that imposes a lower cognitive burden on the end-user. While explanations are helpful, presenting Shapley plots for every individual documentation instance would impose an excessive cognitive burden, as these plots convey highly detailed information.

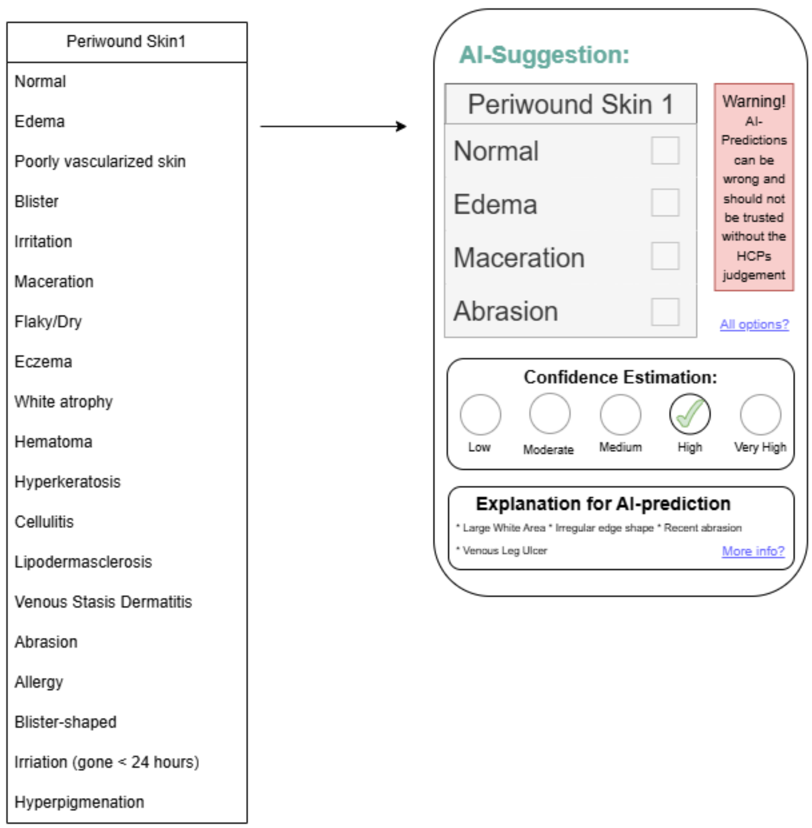

Figure 1: Illustration of phone-based documentation interface replacing a dropdown menu

3.3. Experimental Setup

Each target variable was modeled using a random forest classifier from the `scikit-learn` library. The data were partitioned into ten folds in such a way that every wound appeared in the test set exactly once across the ten evaluation runs. The remaining 9 folds were aggregated and split again for training, validation, calibration of the conformal predictor and calibration of Venn-Abers predictors. All visits associated with a given wound were assigned to the same data split to prevent information leakage arising from repeated measurements. Furthermore, to preserve exchangeability within the calibration set, a single visit was randomly selected from each wound, thereby avoiding dependencies introduced by multiple observations from the same wound. Alternative approaches, such as clustered or hierarchical conformal prediction, have been proposed to account for dependence structures arising from repeated measurements while retaining all available observations Dunn et al. (2023). Classes with fewer than 100 observations were omitted from the study. The performance of the conformal predictor was compared against three benchmarks; APS, RAPS, and a naive approach which achieves the same statistical guarantee by selecting the top-k labels such that their aggregated empirical frequency in the training set and calibration set exceeds $1 - \varepsilon$.

The five target variables are correlated, and information from one variable can be informative for predicting another. To leverage these dependencies, we propose a modeling workflow in which each classifier incorporates the previously documented variables as an additional set of input features as illustrated in Figure 2. The workflow was designed with a practical implementation in mind, assuming that clinicians go through an EHR system where they document variables in a pre-defined order. A real-world implementation of the framework would require a discussion with clinicians to judge whether this assumption holds and a backup plan for how to deploy models when clinicians deviate from the expected documentation order. The implementation of the scheme in Figure 2 followed the same ordering of variables as the columns in Table 1; Tissue Type, Periwound Skin 1, Periwound Skin 2, Edge 1 and Edge 2. All results will be presented in this order.

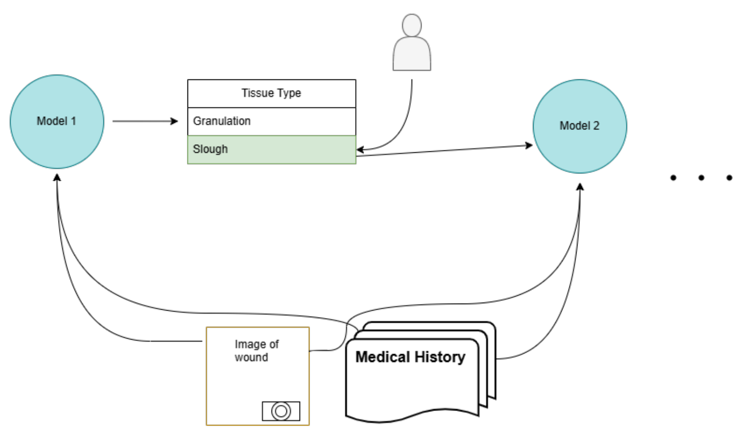

Figure 2: Illustration of workflow in documentation that incorporates newly documented variables as features for the next prediction.

It is important to note that the coverage guarantee applies to each prediction task individually and does not extend to the entire documentation workflow. Achieving a workflow-level guarantee would require a multiple-testing adjustment, such as a Bonferroni correction, across the five prediction tasks. However, this would result in substantially more conservative prediction sets and larger set sizes. Moreover, since the documentation process is inherently performed one variable at a time, a joint workflow-level guarantee may offer limited practical benefit while making the uncertainty estimates more difficult for clinicians to interpret and act upon. Furthermore, several clinical and image-derived features exhibit substantial correlations. As standard Kernel SHAP does not explicitly account for fea-

	$\varepsilon = 0.02$ (98% Coverage)					$\varepsilon = 0.05$ (95% Coverage)				
Problem	Metric	CP	Naive	APS	RAPS	Metric	CP	Naive	APS	RAPS
Tissue Type	Size	1.892	4.0	2.529	2.681	Size	1.377	3.0	1.937	2.021
	Error	0.021	0.018	0.009	0.008	Error	0.052	0.035	0.020	0.018
Periwound Skin 1	Size	7.919	12.0	8.398	8.370	Size	4.721	8.0	5.845	6.354
	Error	0.019	0.015	0.016	0.016	Error	0.048	0.047	0.034	0.031
Periwound Skin 2	Size	3.687	8.0	4.772	4.770	Size	1.576	4.0	2.882	2.951
	Error	0.020	0.016	0.014	0.015	Error	0.049	0.040	0.025	0.028
Edge 1	Size	7.620	11.0	8.695	8.752	Size	4.665	9.0	5.696	6.238
	Error	0.020	0.016	0.014	0.014	Error	0.049	0.037	0.035	0.031
Edge 2	Size	3.897	7.8	4.748	4.721	Size	1.237	4.0	2.518	2.655
	Error	0.021	0.017	0.016	0.017	Error	0.052	0.042	0.031	0.031

	$\varepsilon = 0.10$ (90% Coverage)					$\varepsilon = 0.20$ (80% Coverage)				
Problem	Metric	CP	Naive	APS	RAPS	Metric	CP	Naive	APS	RAPS
Tissue Type	Size	1.106	2.7	1.593	1.603	Size	0.894	2.0	1.301	1.380
	Error	0.103	0.060	0.036	0.035	Error	0.201	0.104	0.064	0.055
Periwound Skin 1	Size	3.120	5.0	3.942	4.216	Size	1.926	4.0	2.642	2.703
	Error	0.100	0.085	0.071	0.064	Error	0.199	0.137	0.135	0.130
Periwound Skin 2	Size	1.062	2.0	1.849	1.842	Size	0.875	1.0	1.318	1.333
	Error	0.098	0.083	0.042	0.045	Error	0.190	0.117	0.066	0.066
Edge 1	Size	3.042	7.0	3.944	4.215	Size	1.812	5.0	2.499	2.737
	Error	0.098	0.085	0.068	0.063	Error	0.195	0.184	0.135	0.125
Edge 2	Size	0.966	1.0	1.469	1.527	Size	0.848	1.0	1.081	1.089
	Error	0.097	0.073	0.046	0.045	Error	0.195	0.073	0.061	0.060

Table 2: Prediction set sizes and empirical errors for all problems at four target error levels, averaged over 10 folds per experiment.

ture dependencies when estimating feature contributions, the resulting attributions may be sensitive to how importance is distributed among correlated variables. Consequently, the explanations should be interpreted as approximate descriptions of model behaviour rather than definitive or causal explanations of individual predictions.

4. Results

This section presents the results of the paper. It first shows the reduction of documentation alternatives with conformal prediction across different ε values and different classification tasks. Then it evaluates the calibratedness of the prediction probabilities both in general and in terms of the five bins from Figure 1. Further, Shapley plots are shown to visualize the explanations of set predictions as feature value contributions.

In Table 2, the average set size produced by conformal prediction across 10 folds for four values of ε across the five classification tasks is presented and compared against APS, RAPS and the naive approach described in 3.3.

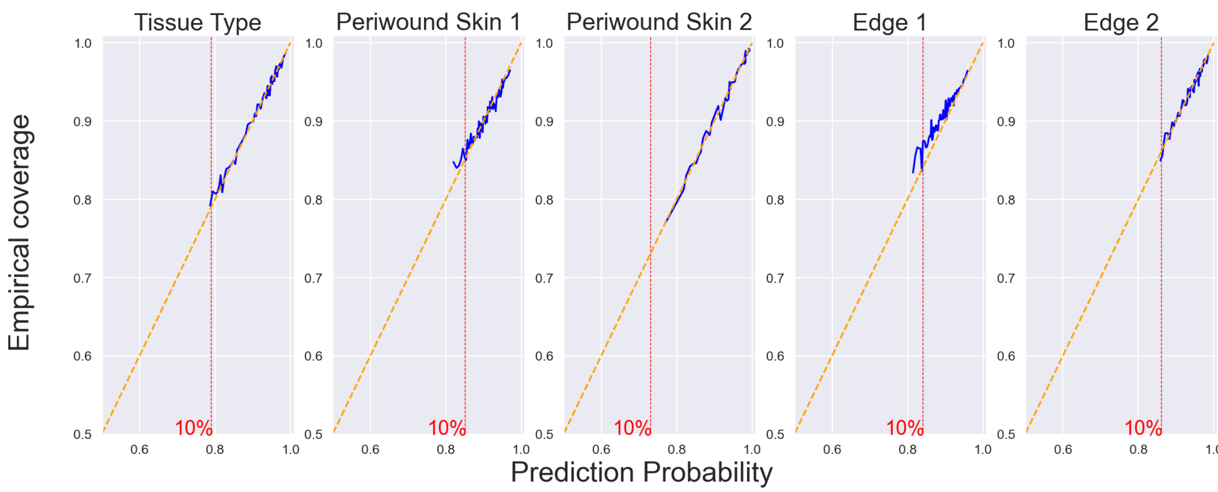

Figure 3: Calibration plot across the 5 classification problems. The blue line shows aggregate coverage and average prediction probability across bins of size 1000. Bins where the difference between the largest and the smallest prediction probability was more than 1% were discarded. The orange line is the line of perfect calibration.

Table 2 shows that conformal prediction reduces the documentation alternatives more than APS and RAPS, and considerably more than the naive approach, across each prediction problem and value of ε. In predicting Periwound Skin 1 and Edge 1, the model generally shows more uncertainty. For the remaining results, $\varepsilon = 0.1$ was chosen for Tissue Type, Periwound Skin 1, and Edge 1, as it provides a favorable trade-off between coverage and prediction set size. For Edge 2 and Periwound Skin 2, however, this significance level occasionally produces empty prediction sets, which are of limited practical utility in a clinical documentation setting. Therefore, a more conservative choice of $\varepsilon = 0.05$ was used for these tasks instead. Figure 3 shows the empirical coverage across bins of 2000 examples ordered by their respective prediction probabilities. The y-axis shows the empirical coverage for each bin.

Figure 3 illustrates close agreement between prediction probabilities of the label being in the set and empirical coverage. This trend holds across bins and prediction tasks, indicating that the model's probability estimates are reliable and well-calibrated. For Edge 1, there is a small underestimation of the prediction sets. Figure 4 shows the same comparison but with prediction probabilities for all five problems being accumulated and the bins being reduced to five bins corresponding to the confidence indicators in Figure 1.

Figure 4 illustrates that the method can provide binned confidence estimations, as those shown in Figure 1, whose coverage is within the prediction interval for the given bin. This is the same observation, only less granular, as made in Figure 3. Because the expected value of coverage over the whole dataset before an example is observed is the coverage probability from conformal prediction, it is expected that values closer to $1-\varepsilon$ are more common. This most likely explains why, within each bin, there is a tilt towards the marginal coverage probability. This leads to the black line being closer to the orange line in the lower bins,

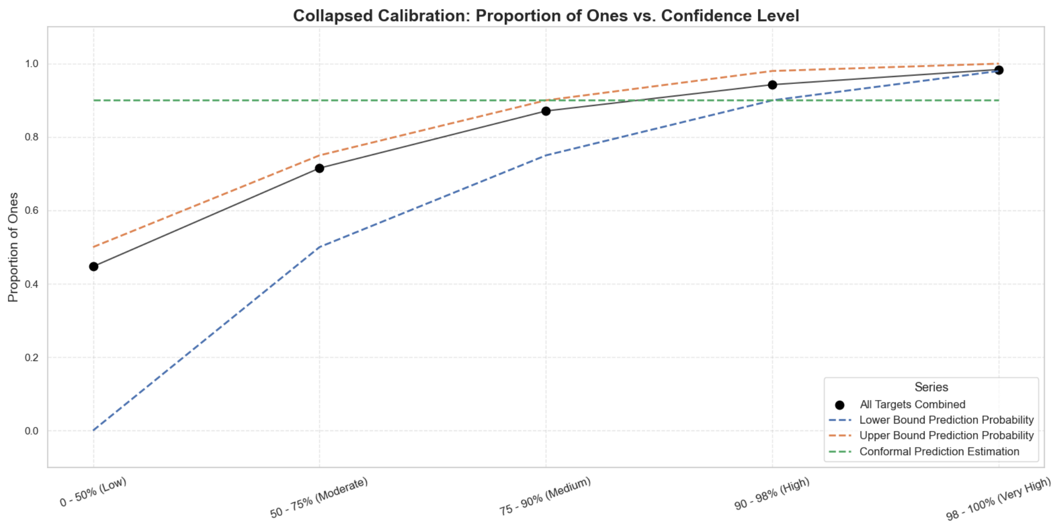

Figure 4: Coverage across all 5 target variables combined within each of the 5 bins presented in Figure 1. The blue and orange line show the Venn-Abers prediction probabilities while the green line shows the coverage information provided by conformal prediction alone.

Problem	Average Diff	Proportion > 0.01	Proportion > 0.05	Proportion > 0.1
Tissue Type	0.0041	0.0677	0.0053	0.0022
Periwound Skin 1	0.0020	0.0128	0.0020	0.0010
Periwound Skin 2	0.0013	0.0004	0.0000	0.0000
Edge 1	0.0023	0.0147	0.0022	0.0006
Edge 2	0.0009	0.0005	0.0000	0.0000

Table 3: Average difference between p_0 and p_1, and the proportion of times the difference is larger than 0.01, 0.05, and 0.1.

and closer to the blue line in the highest bin. This is not a feature of the method but rather a feature of the distribution within any given bin.

As described in Section 2.3, a narrow interval displays confidence in the probability estimate of the Venn-Abers. Table 3 shows the average difference between p_0 and p_1, as well as the number of times that value exceeds 0.01.

4.1. Explanations for set predictions

Below, four waterfall SHAP plots are presented. The information shown in these plots was deliberately selected to maintain patient privacy. The displayed feature values represent high-level, non-identifying variables such as the sequence of recent tissue types and the proportions of pixel-level colour categories within the tissue, edge, or periwound regions. As noted previously, these explanations should not be interpreted as definitive causal explanations of the model's predictions. Rather, they provide an interpretable representation of how individual features contributed to a given prediction according to the trained model, including both the magnitude and direction of their influence. Given the presence of correla-

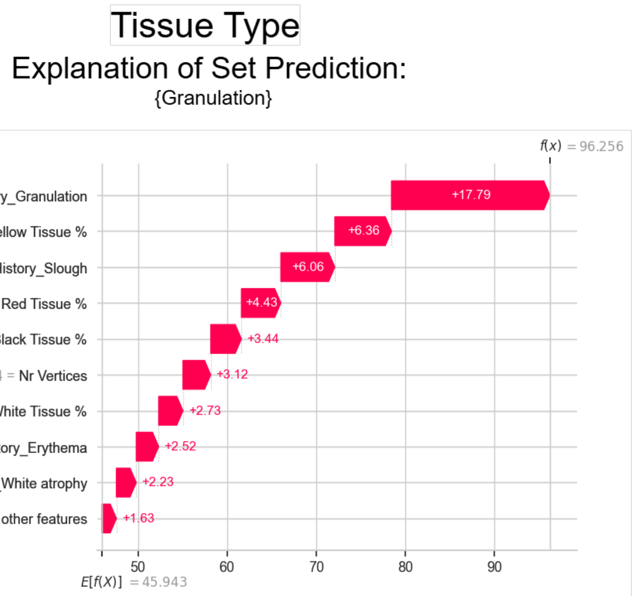

Figure 5: Waterfall plot for the prediction set: {Granulation}

tions among input features, the attributions should be viewed as descriptive and exploratory indicators of model behaviour rather than uniquely correct explanations of the underlying prediction. The examples presented here are intended primarily as illustrative case studies demonstrating how model predictions can be decomposed into feature-level contributions. A more comprehensive assessment of explanation quality would require additional evaluation of attribution faithfulness, such as perturbation- or ablation-based analyses, which are beyond the scope of the present work.

The waterfall plots display the impact of the most important features for explaining the model's confidence in the set prediction relative to a baseline confidence in the corresponding label of a sample of other datapoints being in the set. The results are connected to real-world clinical guidelines when possible. Note that the confidence associated with a set prediction may arise as much from the absence of evidence supporting labels outside the set as from the presence of evidence supporting the labels within it.

The waterfall plot in Figure 5 shows that a long history of documented granulation increases the confidence that the current tissue type is granulation. In addition, 48.6 % red tissue increases the confidence even more. Red tissue is an indicator of granulated tissue Praktisk Medicin (2024). A lack of history of slough, the second most common tissue type also increases the confidence. A low proportion of white and yellow tissue also increases the confidence, these colours are indicators of slough European Wound Management Association (2014). Similarly, a lack of black tissue increases confidence, black tissue indicates necrotic tissue Praktisk Medicin (2024). All in all, the feature values increase the confidence from the expected value across the dataset of 46% to 96%.

The waterfall plot in Figure 6 shows a value of 1.75 for history of necrotic tissue, indicating that the three most recent visits were labeled as containing necrotic tissue. This history increases the model's confidence in the prediction, whereas the absence of a his-

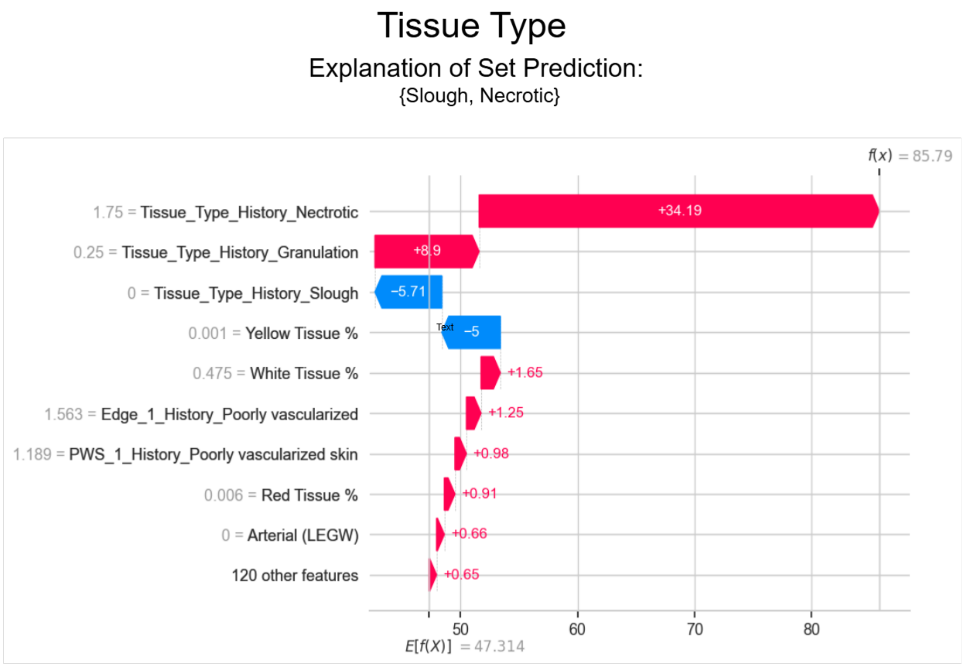

Figure 6: Waterfall plot for the prediction set: {Slough, Necrotic}

tory of slough decreases it. The wound was labeled with granulation tissue four visits ago, which also contributes positively to the prediction. Similarly, the absence of yellow tissue decreases the confidence, while a proportion of 47.5% white tissue increases it. As noted above, both white and yellow tissue are indicators of slough. In total, the feature values increase the confidence from the expected value across the dataset of 47% to 86%.

Figure 7 shows that the wound being on the foot decreases the confidence in the prediction set. Table 1 shows that callus, which is typically on the feet, is the most common label not included in the prediction set in Figure 7. The label of the last visit was normal, increasing the confidence in the set. The wound has never been labeled with callus, also increasing the confidence in the set. The second to last visit was labeled as maceration, decreasing the confidence in the set. We see that the confidence is 76% on the background sample, which is because it is a prediction set of the four most common labels. The features only increase the confidence with 1 percentage point. This indicates that the model actually has not found much to go on for this prediction, instead it utilizes the four dominant classes.

Figure 8 shows that a lack of previous observations of Normal as the Periwound Skin label has a negative effect on the confidence in the prediction set. Conversely, a previous history of flakiness/dryness increases the confidence in the prediction. The wound is placed on the feet which increases the confidence in the prediction set. The wound has a ratio $Area/Perimeter^2$ of 0.046, increasing the confidence by almost 1 percentage point. To put this in perspective, a circle, which has the maximum possible ratio, has $\frac{1}{4\pi} \approx 0.079$ and a square has $\frac{1}{16} \approx 0.0625$. There is very little black Periwound Skin which is increasing the confidence of the set prediction. Neither of the five Periwound Skins in the set are typically

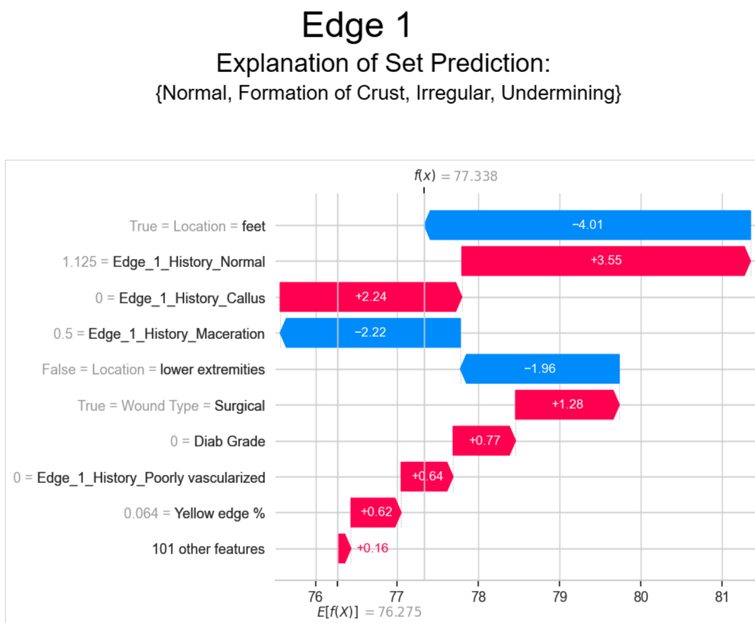

Figure 7: Waterfall plot for the prediction set {Normal, Formation of Crust, Irregular, Undermining}

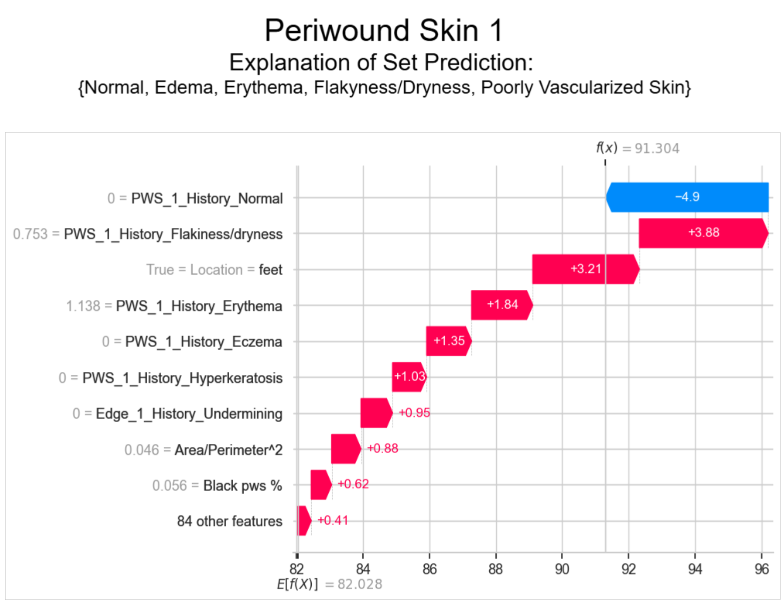

Figure 8: Waterfall plot for the prediction set {Normal, Edema, Erythema, Flakiness/dryness Poorly vascularized skin.}

black. Edema is typically pale or slightly pink, erythema is typically red, dry/flaky skin is typically white or gray, and poor vascularization is typically a blue or purple colour.

It is worth noting that the prediction set contains many of the most common Periwound Skin types and therefore the baseline confidence is already quite high.

5. Discussion

The integration of AI-based tools into healthcare offers substantial opportunities, while also introducing important ethical challenges. Clinical documentation is a natural starting point for adoption, as it carries relatively lower immediate risk while remaining one of the most time-consuming tasks for clinicians. However, even seemingly routine documentation can influence downstream clinical decisions, making reliability essential. In this context, uncertainty quantification becomes a critical mechanism for addressing ethical concerns and strengthening trust in AI-generated outputs.

The full label-space sizes are reported in parentheses in the column headers of Table 1. As shown in Table 2, the reduction in the number of candidate labels relative to the full label space is substantial even for small values of ε. This suggests that the proposed method could considerably reduce the documentation burden for clinicians by limiting the number of alternatives that need to be considered.

Table 2 further demonstrates that all conformal prediction methods produce substantially smaller prediction sets than the naive approach, which is expected given that the underlying model possesses non-trivial predictive power. Among the conformal methods, standard conformal prediction consistently yields smaller prediction sets than APS and RAPS at the same ε level. This difference appears to be largely explained by the fact that APS and RAPS are more conservative, yielding error rates that are often well below the specified tolerance. One plausible explanation for this behavior is the discrete nature of the label space. APS and RAPS rely on cumulative class probabilities, which may induce a relatively coarse distribution of conformity scores compared to standard conformal prediction. As a result, the methods may be unable to precisely attain the target error level ε, yielding conservative coverage and larger prediction sets. This interpretation is supported by the observation that the conservativeness is most pronounced for tissue type prediction, which has the smallest label space among the investigated tasks. A smaller number of classes naturally leads to fewer possible conformity score values, potentially exacerbating discretization effects and making it more difficult to achieve error rates close to the nominal level. In contrast, standard conformal prediction produces error rates that more closely match the target ε, thereby making more efficient use of the permitted error budget. Consequently, the smaller prediction sets obtained by standard conformal prediction should not be interpreted as reflecting a different coverage-error trade-off, but rather as a result of achieving calibration that is closer to the nominal level. Whether exact adherence to the target error level is important in practice, or whether the primary consideration should instead be the resulting trade-off between prediction set size and empirical error rate, is ultimately a question that should be evaluated together with clinicians and other stakeholders.

Naturally, improving the model's predictive accuracy can reduce conformal set sizes even further. Deep learning models such as CNNs may offer such improvements, but explaining their predictions is typically more difficult than for the feature-engineered image representation used in this work. When colour attributes like "yellow" exist as explicit features, their influence is directly visible in SHAP plots, whereas in CNNs such information is entangled in high-dimensional internal representations. This trade-off between predictive power and interpretability must be carefully navigated. The emergence of large language models may also create opportunities for richer explanations when combined with existing techniques such as saliency maps. Another avenue for reducing set sizes is to improve data quality and increase dataset size. The Wound Monitor dataset contains at least 600,000 images, of which 185,000 were used in this work. The subset we rely on consists exclusively of images that include a clinician-annotated polygon. Thus, both improving data quality and increase dataset size are feasible objectives. This needs to be done in a careful manner, likely with frameworks from uncertainty quantification. We aim to pursue these directions in future work.

Table 3 shows that the interval between p_0 and p_1 is typically very small, with mean values below 0.005 for all prediction tasks. However, there are occasional cases in which the difference exceeds each of the three considered thresholds. The practical significance of such differences likely depends not only on their magnitude but also on the underlying probability values. For example, a decrease from 0.99 to 0.94 may be clinically more consequential than a decrease from 0.66 to 0.62, despite the latter representing a similar absolute change. As with many other nuanced aspects of clinical deployment, determining how such uncertainty should be communicated and acted upon will require discussion with clinicians and other stakeholders. Nevertheless, the present results suggest that some form of uncertainty flag may be warranted for the rare occasions in which the interval between p_0 and p_1 exceeds 0.1.

Figures 3 and 4 show that the prediction probabilities assigned to the sets are trustworthy. The proportion of times the label is in the set corresponds to the prediction probabilities. This means they can be used to inform clinicians about the trustworthiness of a set prediction and thus reduce the risk that the speed-up of the documentation process compromises the quality of the documentation. However, exactly how they should be used remains an open question, there needs to be a balancing of the objective of reducing clinicians' cognitive burden while still conveying the information essential for sound decision-making. In this article, we have presented one approach with the binning in Figure 1, but there are alternative approaches. One plausible approach is to treat these probabilities as a reject option, withholding any set whose predicted probability falls below a predefined threshold—for instance, 50%, corresponding to the low-confidence bin. If confidence bins are the way to go, they should be designed in a less arbitrary fashion, potentially adaptive to the documentation task. 98% might be considered high confidence for a Periwound Skin type but not for a more severe condition. Similarly, whether the workflow presented in Figure 2 reduces the practicality of the solution is most likely context dependent and depends on contextual factor such as the IT-systems in place. Addressing these questions requires direct input from clinicians and administrative workers in healthcare, whose expertise and lived experience shape both the practical constraints and the expectations for any AI-supported workflow.

Interpreting explanations for set predictions can be more challenging than interpreting explanations for individual labels. As shown in Figures 5, 6, 7 and 8 , however, it is certainly possible to make sense of the feature contributions. The key is to frame the explanation around the binary question that the confidence score actually addresses: *Is the true label contained in the predicted set?* Viewed this way, the distributional information in Table 1 becomes directly relevant. Across all Figures, the histories of the most common labels outside the set influence the confidence, sometimes even to the same degree as the histories of labels inside the set. This is expected, since understanding why a model assigns confidence to a particular outcome is inherently equivalent to understanding why it does not assign confidence to the complementary outcome. Even though one can adapt to interpreting these SHAP plots, the added value of a set-level explanation—compared to explanations for individual labels—may not be immediately obvious. We argue that the value is two-fold. First, showing how multiple members of a set are jointly pushed in the same direction by particular features provides a coherent rationale for trusting the set as a whole. When the precise label is less important than membership in an actionable superclass, this abstraction level is the most meaningful one for decision-making. Second, set-level explanations display feature contributions at the level at which they actually affect the confidence in the set. Individual SHAP plots obscure this, because—unless additional procedures are applied—each label-specific plot is implicitly treated as equally influential, even though some labels may contribute far more to the model's confidence. A one-against-all approach combined with the proposed confidence-assignment framework partially mitigates this issue, since the actual percentage contributions are visible in the plots. However, this approach fails to account for dependency structures among labels, as discussed in Section 2.5.1. Using compositional Shapley values resolves the dependency issue, but at the cost of losing information about the relative magnitudes of label influence, since the values are computed in a high-dimensional simplex. Set-level SHAP explanations have neither of these problems, but as discussed, might explain things on the wrong level of abstraction if one is only concerned with individual label explanations. The optimal approach is thus highly context-dependent.

It is also important to acknowledge ethical concerns that fall outside the technical scope. As AI adoption becomes more widespread, over-reliance on automated systems represents a persistent risk. Furthermore, the concerns expressed by clinicians in municipal home healthcare regarding the relational and embodied nature of wound care constitutes another risk. Any implementation of the proposed framework should therefore be carried out with careful attention to clinicians' established workflows, aiming to support efficiency gains without undermining professional autonomy and practices. Still, we argue that the approach presented in this work helps mediate these concerns because it does not aim for full automation - the clinician will always make the final call. Instead, it focuses on filtering out unlikely documentation alternatives while maintaining transparency around uncertainty and reasoning, which directly addresses several of the core issues raised by clinicians.

Finally, a natural extension of the method is to enlarge the set by adding additional candidate labels until the prediction probability exceeds a desired threshold. However, this strategy introduces multiple-testing concerns that must be carefully addressed. We aim to explore this direction in future work.

6. Concluding remarks

Addressing the burden of clinical documentation is essential for the well-being of both clinicians and patients. The adoption of AI in this domain requires frameworks that support trustworthy and transparent predictions. In this work, we have shown how conformal prediction, Venn–Abers predictors, and SHAP can be combined into an integrated framework that produces prediction sets which meaningfully reduce documentation alternatives while providing well-calibrated probabilities—and corresponding explanations—of how likely the set is to contain the true label. This framework can reduce clinicians' documentation time in a trustworthy way that preserves their autonomy and established practices.

7. Code and Data availability

The study was approved by the Swedish Ethical Review Authority (reference no. 2024-07554-01). Due to restrictions imposed by the approved ethical review protocol, the data used in this study cannot be shared publicly. While the data and study-specific implementation are not publicly available, the underlying framework used in this work is available at https://github.com/OskarGustafsson123/CP-VA.

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